

Power Control Programming Manual

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Overview

The triggered USB module provides 0-10V, current limited to 200mA. A resistor can be changed to increase the current to several amps, though there probably isn't enough heat sinking to provide that much current for low voltages. Until the output voltage is updated, the output is disabled and leaks ~-3mA. The voltage can be set by commands sent over USB using an FTDI chip to communicate. Voltage levels can be stored in non-volatile memory that will output based on the memory address. The memory address is advanced one step for each Step trigger signal received. The address is reset to the start address when a Sweep trigger signal is received or when the "Set Start Address" command is sent. The step and sweep triggers are 0-3.3 V or 0-5 V, active on the transition low to high. The "Set Start Address" command sets the start address and initiates the module to allow stepping with trigger signals. All other commands stop the module from responding to trigger signals. Use Set Start Address to initiate sweeps, and use Set Voltage to interrupt sweeps. This document contains instructions for using the commands sent over the USB.

Command and Return Structure

Commands can be sent and data received by a computer as a series of bytes by using FTDI device drivers or by using the virtual COM port. The FTDI device drivers must be installed before connecting the USB module to a computer. After the USB module is connected to the computer, the device can be accessed through a COM port, or the bytes can be sent using the function "FT_Write" in the device driver FTD2XX.dll, and the return bytes can be read with "FT_Read". The command structure uses one command byte followed by data.

Commands

Set Voltage (0x56, ASCII: V, returns 0 bytes):

The output voltage can be set by sending 3 bytes over the USB. The first byte is the command byte 0x56, followed by two unsigned bytes representing the voltage, MSB.

Send:

BYTE1: Voltage command, 0x56.
BYTE2: Most significant byte of the 16 bit voltage.
BYTE3: Least significant byte of the 16 bit voltage.

The two bytes representing the voltage can be obtained by the following formula:

$$V_{\text{bytes}} = V_{\text{set}} / 10 * 65536$$

where V_{set} is the desired output voltage. If V_{bytes} is greater than 65535, 65535 will be used to set the output voltage to the maximum. All three bytes must be sent even if one of the bytes is zero.

Following is an example sequence in hexadecimal format to set the voltage to 5V.

0x56, 0x80, 0x00.

The output voltage ranges from 0-10V. This command also disables the trigger response.

Load Memory (0x57, ASCII: W, returns 0 bytes):

The onboard memory contains 524,288 16-bit words. Each 16-bit word represents a voltage according to the formula in the Set Voltage command. The memory must be written after each power cycle and is written as a series in order of ascending addresses. The series can begin at any address from 0-524,287 and contain up to 2^{32} entries, though after the last memory is written, it will cycle back to address 0 and potentially overwrite previous data. The first byte is the command byte 0x57, followed by 4 unsigned bytes representing the number of 16 bit words, 4 bytes representing the starting address, and 2 bytes per word repeated for the total number of words, all MSB. The number of 16 bit words is the number of voltage settings placed in memory.

Send:

BYTE1: Power command, 0x57.
BYTE2: Most significant byte of the 32 bit number of voltage settings.

BYTE3: Next significant byte of the 32 bit number of voltage settings.
 BYTE4: Next significant byte of the 32 bit number of voltage settings.
 BYTE5: Least significant byte of the 32 bit number of voltage settings.
 BYTE6: Most significant byte of the 32 bit start address.
 BYTE7: Next significant byte of the 32 bit start address.
 BYTE8: Next significant byte of the 32 bit start address.
 BYTE9: Least significant byte of the 32 bit start address.
 BYTE10: Most significant byte of the 16 bit voltage.
 BYTE11: Least significant byte of the 16 bit voltage.
 BYTE12: Most significant byte of the next 16 bit voltage.
 BYTE13: Least significant byte of the next 16 bit voltage.
 BYTEN: Most significant byte repeated for each 16 bit voltage written
 BYTE N+1: Least significant byte repeated for each 16 bit voltage written

For example, to write 2 memory locations starting at address 3 with 0V and 10V do

0x57, 0x00, 0x00, 0x00, 0x02, 0x00, 0x00, 0x00, 0x03, 0x00, 0x00, 0xff, 0xff.

Set Start Address (0x53, ASCII: S, returns 0 bytes):

When the module receives a sweep trigger low to high transition, the start address for the output sweep is reset to the value provided by this command. The first byte is the command byte 0x33, followed by 4 unsigned bytes representing the start address, all MSB.

Send:

BYTE1: Set start address command, 0x33.
 BYTE2: Most significant byte of the 32 bit start address.
 BYTE3: Next significant byte of the 32 bit start address.
 BYTE4: Next significant byte of the 32 bit start address.
 BYTE5: Least significant byte of the 32 bit start address.

The sweep is reset to this address after the module receives this command and will begin sweeping as soon as it receives a trigger signal. When the module receives the sweep trigger, the address will be reset to this starting address.

The next set of commands are for diagnostic purposes:

Output Memory (0x40, ASCII: @, returns 0 bytes):

This command allows a series of memory locations to be converted to the voltage output without requiring any trigger signals by specifying the number of memory locations and the starting address. After receiving the command, the output will cycle through those memory locations in ascending order beginning at the starting address. The step rate is about 11uS and cannot be changed. Once the module steps through the number of addresses specified by this command, the sweep will stop. This command does not change the starting address used for the triggered sweep, though it will disable the trigger response, and Set Starting Address will need to be sent

to restart the triggered sweep. The first byte is the command byte 0x40, followed by 4 unsigned bytes representing the number of addresses to step through and 4 bytes representing the starting address, all MSB.

Send:

BYTE1: Output memory command, 0x40.
BYTE2: Most significant byte of the 32 bit number of addresses.
BYTE3: Next significant byte of the 32 bit number of addresses.
BYTE4: Next significant byte of the 32 bit number of addresses.
BYTE5: Least significant byte of the 32 bit number of addresses.
BYTE6: Most significant byte of the 32 bit start address.
BYTE7: Next significant byte of the 32 bit start address.
BYTE8: Next significant byte of the 32 bit start address.
BYTE9: Least significant byte of the 32 bit start address.

For example, to have the USB Module start at address 5 and step through 8 memory locations:

0x40, 0x00, 0x00, 0x00, 0x08, 0x00, 0x00, 0x00, 0x05.

Read Back Memory (0x52, ASCII: R, returns N bytes):

This command reads back the memory from the board. Specify the number of addresses to read back and the starting memory, and the USB Module will send 2 bytes per memory location. The first byte is the command byte 0x52, followed by 4 unsigned bytes representing the number of addresses and 4 bytes representing the starting address, all MSB.

Send:

BYTE1: Read back command, 0x52.
BYTE2: Most significant byte of the 32 bit number of addresses.
BYTE3: Next significant byte of the 32 bit number of addresses.
BYTE4: Next significant byte of the 32 bit number of addresses.
BYTE5: Least significant byte of the 32 bit number of addresses.
BYTE6: Most significant byte of the 32 bit starting address.
BYTE7: Next significant byte of the 32 bit starting address.
BYTE8: Next significant byte of the 32 bit starting address.
BYTE9: Least significant byte of the 32 bit starting address.

Receive:

BYTE1: Most significant byte of the 16 bit memory value.
BYTE2: Least significant byte of the 16 bit memory value.
BYTEN: Most significant repeated for the number of memory locations specified.
BYTEN+1: Least significant byte repeated for the number of memory locations specified.

For example, to read back 2 memory locations starting at address 4 send,

0x52, 0x00, 0x00, 0x00, 0x02, 0x00, 0x00, 0x00, 0x04.